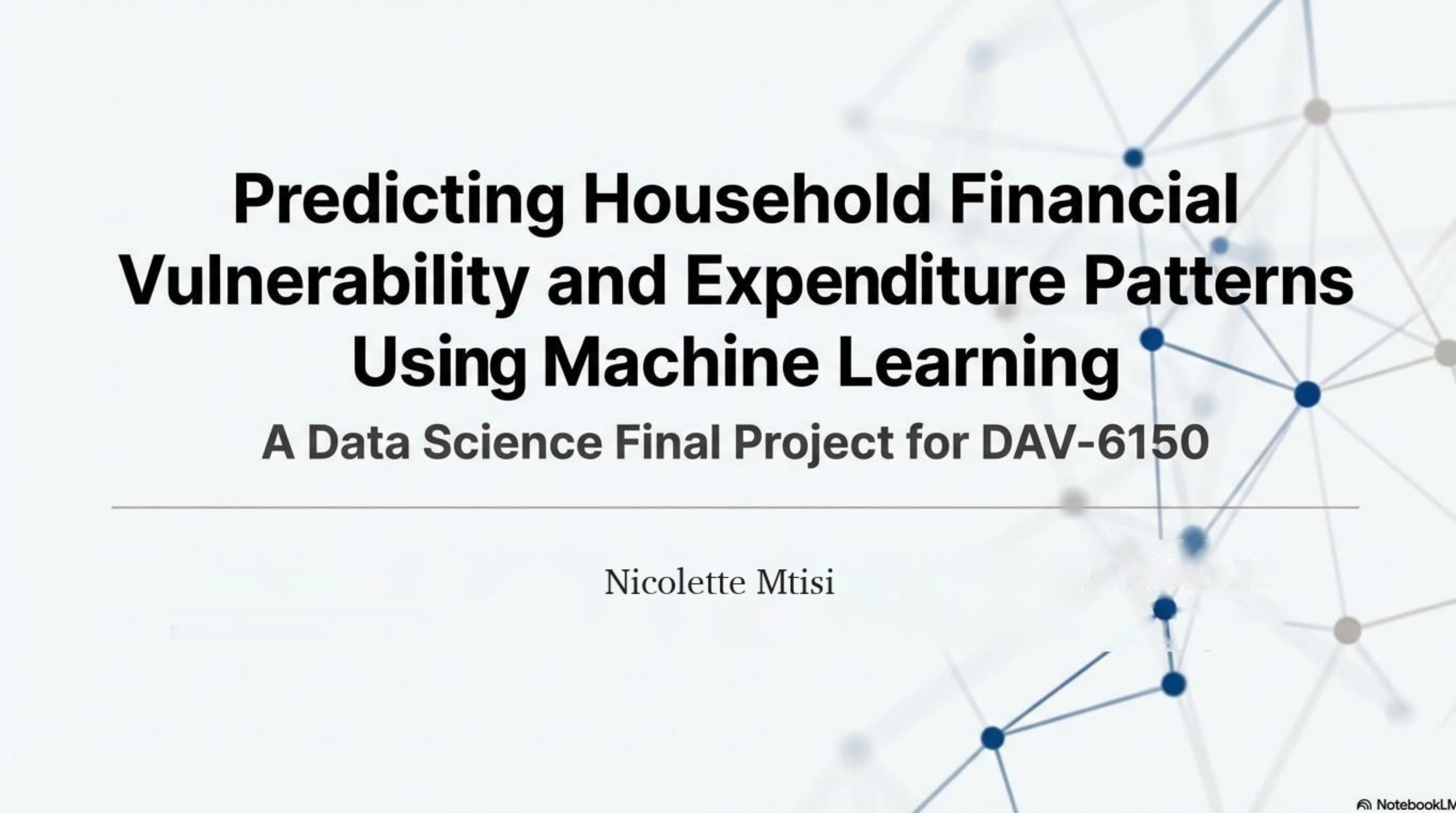




Predicting Household Financial Vulnerability and Expenditure Patterns Using Machine Learning

A Data Science Final Project for DAV-6150

Nicolette Mtisi

A Third of American Households Are on a Financial Knife's Edge

37%



Approximately 37% of American adults would struggle to cover an unexpected \$400 expense using cash or its equivalent.

Source: Federal Reserve, 2024

28%



Nearly three in ten U.S. adults (28%) expect their household's financial situation to worsen within the next year.

Source: Pew Research Center, 2025

Understanding the determinants of financial health is critical for policymakers, financial institutions, and community organizations. Machine learning offers a powerful lens to uncover complex, non-linear patterns that traditional methods may miss.

The Foundation: The U.S. Consumer Expenditure Survey

Our analysis is built on the 2023 Quarter 2 Interview File from the U.S. Bureau of Labor Statistics (BLS). This is a gold-standard, nationally representative dataset, not sourced from Kaggle or other repositories.



Source: U.S. Bureau of Labor Statistics (BLS)



Dataset: `fmli232.csv` (2023 Q2)



Unit of Analysis: Consumer Unit (a household or individuals sharing expenses)



Initial Size: 4,751 households



Initial Features: 819 variables covering demographics, income, expenditures, assets, and geography.

Our Investigation Is Guided by Three Core Research Questions



Classification

Can we accurately predict household financial vulnerability?

Vulnerability is a binary outcome defined by: expenditures > 95% of income, OR housing costs > 50% of income, OR essential spending > 80% of income.



Regression

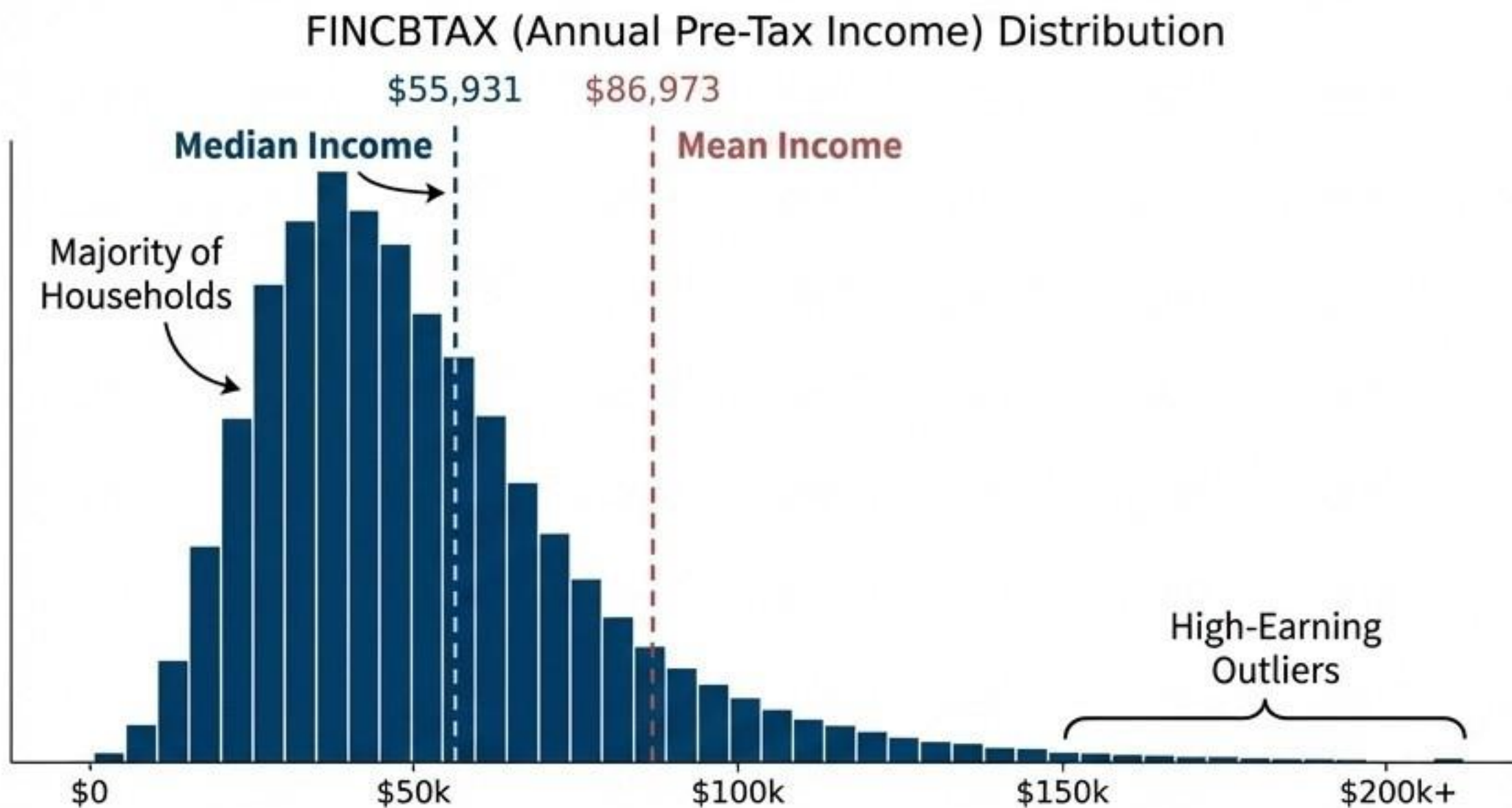
What are the primary drivers of total household expenditure?



Clustering

Can we identify distinct household spending profiles or consumer segments?

EDA Insight: The Income Distribution Reveals a Deep Inequality Gap



An analysis of annual pre-tax income ('FINCBTAX') confirms a classic, highly right-skewed distribution, a hallmark of U.S. economic data.

Implication for Modeling

The extreme skew and presence of outliers necessitate robust data preparation. This justifies our use of **median imputation** for missing values and **Winsorization** (outlier capping) to prevent a small number of households from distorting the model.

Median Income: \$55,931
(A more representative 'typical' household).

Mean Income: \$86,973
(Pulled up significantly by high earners)

⚠ Data Integrity Issue: Minimum income of **-\$82,067** was found, indicating business losses that require special handling.

EDA Insight: High Income Does Not Directly Translate to High Spending

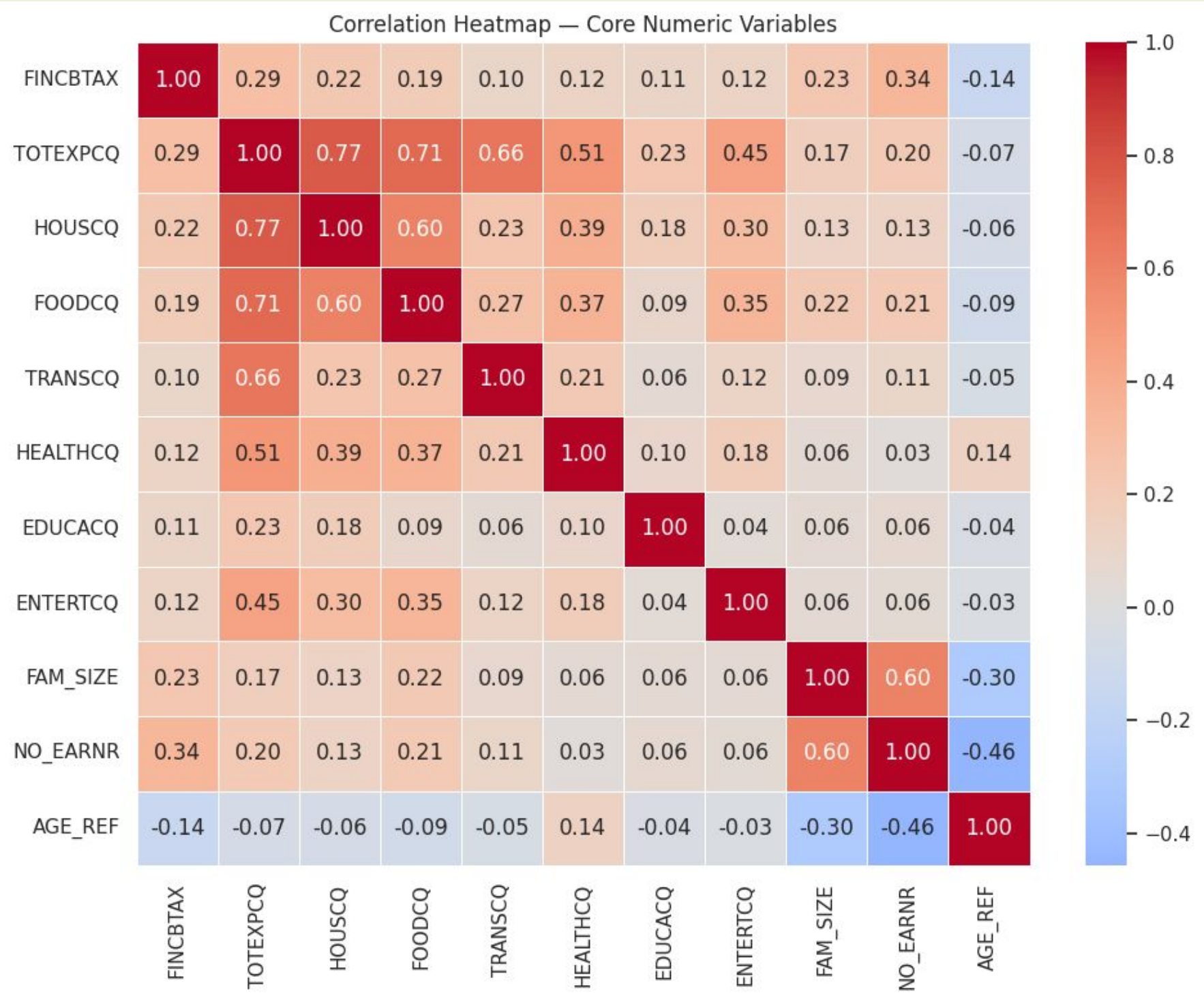
While income and expenditure are positively correlated, the relationship is surprisingly weak (Pearson's $r = 0.29$). This “disconnect” indicates that income is a measure of financial *capacity*, not necessarily financial *behavior*. Many high-income households are savers, not spenders.



Implication for Modeling

The weak linear relationship and fanning-out pattern (heteroscedasticity) confirm that simple linear models will be insufficient. We need **non-linear models** (e.g., Random Forest, XGBoost) capable of capturing these complex interactions and diverse behavioral profiles.

EDA: SPENDING DRIVERS



The strongest red blocks are found in the relationship between Total Expenditure (TOTEXPCQ) and specific spending categories.

- Housing (HOUSCQ) vs. Total Expenditure (0.77): This is the strongest correlation, indicating that housing costs are the primary driver of a household's total budget.
- Food (FOODCQ) vs. Total Expenditure (0.71): Food costs also track very closely with overall spending.
- Transportation (TRANSCQ) vs. Total Expenditure (0.66): This shows a moderate-to-strong link.
- Health (HEALTHCQ) vs. Total Expenditure (0.51): Interestingly, healthcare has a weaker correlation with total spending compared to housing or food, likely because healthcare costs can be sporadic or fixed (insurance) rather than consumption-based.

Housing: Dominant spending across all brackets.

Family Size: Direct linear driver of baseline costs.

Stability: Income source (wages vs fixed) matters more than size.

Strategic Implications for Modeling

EDA insights confirm that the identified feature groups provide the necessary context to move beyond simple income-based vulnerability detection.



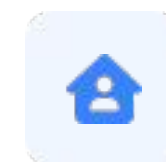
Income Sources

Acts as the baseline measure of **financial capacity**. Modeling income types (e.g., salary vs. welfare) helps distinguish between stable and volatile households.



Expenditure Groups

Housing, Food, and Transport serve as critical **stress proxies**. These non-discretionary costs define the floor of economic resilience.



Contextual Factors

Household Size, Region, and Number of Earners provide the data **normalization** required to interpret spending across diverse demographics.

MODEL STRATEGY

By integrating these three distinct feature groups, we can effectively model **behavioral spending profiles** and identify financial vulnerability with significantly higher nuance than static income analysis alone.

Data Preparation Pipeline

1. Data Cleaning

- Median/mode imputation for missing values
- Outlier capping for income/expenditure

2. Feature Engineering

- Created vulnerability binary target
- Generated income/expenditure ratios

3. Feature Selection

- Removed correlated features ($r > 0.9$)

Encoding Strategy

One-Hot Encoding

Nominal variables

Data Split

70% / 15% / 15%

All preprocessing steps implemented with scikit-learn pipelines for reproducibility

DATA PREPARATION STRATEGY

95% REDUCTION

Removed variables that had more than 95% missing data

IMPUTATION

Applied Median imputation for skewed numeric variables and Mode for categorical demographic gaps.

OUTLIER CONTROL

Winsorized financial variables at 1st and 99th percentiles to stabilize model training.

SUMMARY OF DATA PREPARATION

Transforming raw Consumer Expenditure Survey data into high-integrity datasets optimized for machine learning through rigorous cleaning, engineering, and standardization.



Sparsity Reduction

Applied a strict **80% missingness threshold** for financial assets. Pruned nine low-signal variables to reduce dimensionality and prevent overfitting on empty noise.



Outlier Management

Winsorization: Capped money-like variables at the **1st and 99th percentiles** to neutralize extreme values while preserving total sample size.



Standardization

Continuous variables standardized to $\mu = 0, \sigma = 1$ via Z-score scaling. Applied One-Hot Encoding to all categorical features.



Imputation Strategy

Stabilized distributions via **Median Imputation** for skewed financial data and **Mode Imputation** for demographics, mitigating bias from wealth outliers.



Final Datasets

Produced three leakage-free sets: **X_cls/y_cls** (Classification), **X_reg/y_reg** (Regression), and **df_clust** (Behavioral Clustering).

Integrity Check: All matrices are fully dense (0 missing values) and structurally aligned for training.

Model Inventory & Selection Strategy

Comprehensive technical architecture for RQ1 Classification, RQ2 Regression, and RQ3 Clustering.

RQ1: Predict Vulnerability

Category	Model Architecture	Type
Baseline	LogReg (Balanced)	Linear
Advanced	SVM (Linear / RBF)	Margin / Nonlinear
Tree-Based	XGBoost Classifier	Gradient Boosting
DL	MLP Neural Network	Pattern Recognition

OPTIMAL METRIC: PR-AUC

RQ2: Predict Expenditure

Category	Model Architecture	Type
Baseline	Ridge Regression	Linear (L2)
Selection	LASSO Regression	Linear (L1)
Ensemble	Random Forest	Averaged Trees
Boosting	XGBoost Regressor	Gradient Boosting

OPTIMAL METRIC: Lowest RMSE

RQ3: Household Profiles

Algorithm	Input Features	Final Choice
K-Means	Expenditure Shares	k = 7 Clusters

Selection Criteria: Highest Silhouette Score (0.388) indicating optimal cluster separation and internal cohesion.

METRIC: Silhouette Score

Ensemble Learning (RQ1)

Strategy	Learners	Key Strength
Soft Voting	LR, KNN, SVM, DT	Balanced P-R
Hard Voting	LR, KNN, SVM, DT	Highest F1-score
Stacking	Base → Meta (LR)	Meta-learning

FOCUS: F1-Score & PR-AUC

RQ1: Model Selection

Classification Models Ranked by PR-AUC

Model	Threshold	ROC-AUC	PR-AUC	Precision	Recall	F1
XGBoost ⭐	0.950	0.994	0.952	1.000	0.826	0.905
SVM Linear	-0.006	0.979	0.644	0.500	0.696	0.582
LogReg (balanced)	0.650	0.969	0.497	0.432	0.696	0.533
SVM RBF	0.100	0.919	0.472	0.224	0.652	0.333
MLP	0.050	0.935	0.448	0.520	0.565	0.542

Best Model: XGBoost
Highest PR-AUC (0.952) - optimal for rare-class performance

Key Observations
All models show high ROC-AUC, but XGBoost significantly outperforms in PR-AUC

Models sorted by PR-AUC (Precision-Recall Area Under Curve), then ROC-AUC

RQ2 Model Performance Analysis

Comparative evaluation of regression models for predicting Total Household Expenditure.

Model Name	R ² Score	MAE (USD)	RMSE (USD)
XGBoost WINNER	0.9683	480.48	1300.17
Random Forest	0.9492	665.87	1648.40
Ridge Regression	0.9394	897.17	1800.16
LASSO Regression	0.9392	896.42	1802.90



XGBoost was selected as the optimal model for RQ2. It achieved the **lowest RMSE**, indicating superior overall predictive accuracy and the best performance in handling complex, non-linear spending variance.

RQ3 Clustering Optimization

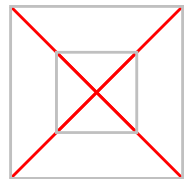
Comparative analysis of K-Means stability using Silhouette Scores across cluster counts.



K-Means with k=7 was selected for the final behavioral profiling. It yielded the highest silhouette score (0.388), indicating the best balance of internal cluster cohesion and distinct separation between segments.

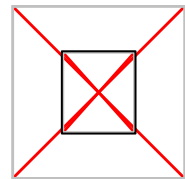
Ensemble Modeling Framework

A comprehensive architecture for RQ1 classification, combining diverse weak learners with advanced aggregation strategies to maximize predictive resilience.



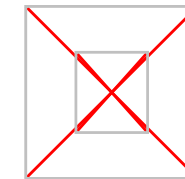
Individual Classifiers

- **Logistic Regression:** Baseline linear probabilistic model.
- **K-Nearest Neighbors:** Instance-based non-parametric learning.
- **Decision Trees:** Captures non-linearities and interactions.
- **Support Vector (SVC):** Complex decision boundary optimization.



Ensemble Methods

- **Voting Classifier:** Aggregates predictions via majority (Hard) or probability (Soft) weights.
- **Stacking Classifier:** Advanced multi-layer architecture using a meta-model to learn optimal combinations.



Evaluation Metrics

- **PR-AUC / ROC-AUC:** Critical assessment for imbalanced class dynamics.
- **Confusion Matrix:** Mapping true vs. false identification of at-risk households.
- **Classification Report:** Precision, Recall, and F1-Score analysis.

Ensemble Learning Performance

Evaluating Stacking and Voting methods for maximizing Financial Vulnerability detection.

Model Strategy	ROC-AUC	PR-AUC	Precision	Recall	F1-Score
Stacking Ensemble	0.9832	0.7327	0.370	0.870	0.519
Soft Voting (Best Balance)	0.9833	0.7264	0.714	0.652	0.682
Hard Voting	N/A	N/A	0.714	0.652	0.682



Efficiency Trade-off

Soft Voting provides a superior balance of identification (F1: 0.68), significantly reducing false alarms with only 6 false positives (Precision 0.71).

The First Challenge: From 819 Variables of Chaos to 44 Signals of Clarity

The raw dataset is characteristic of government surveys—comprehensive but noisy.

Our first task was to systematically isolate the signal by removing administrative, non-predictive, and redundant variables. This reduced the feature space by 95% without losing a single household observation.



A Transparent Audit of Our Variable Reduction

The reduction from 819 to 44 variables was not arbitrary. It was a defensible, rule-based process designed to optimize the signal-to-noise ratio.

Excluded Variables (Noise & Administration)	Retained Variables (Analytical Signal)
✗ 69 Detailed Expenditure Subcomponents (to avoid multicollinearity)	✓ 16 Income Variables (wages, social security, etc.)
✗ 44 Replicate Weights (sampling artifacts, not behavioral data)	✓ 9 Demographic & Household Structure Features (age, family size, etc.)
✗ 13 Administrative & Survey Mechanics (interview flags, metadata)	✓ 8 Major Expenditure Aggregates (total, housing, food, etc.)
	✓ 6 Financial Resilience Indicators (liquidity, debt, assets)
	✓ 4 Geographic Indicators (region, urban/rural)
	✓ 2 Structural IDs (household ID, survey weight)

Based on the Variable Reduction Audit in the project's data preparation phase.

Executive Summary: Financial Behavior Modeling

RESEARCH METHODOLOGIES AND MODEL SELECTION FRAMEWORK

RQ1: CLASSIFICATION

Best: XGBoost

Predicting Household Vulnerability

Models evaluated: Linear, Non-linear Classifiers, Ensembles

- ✓ **Superior Detection:** Highest ROC-AUC (0.994) and PR-AUC (0.952).
- ✓ **Balanced Errors:** Perfect precision (1.00) with strong recall (0.83).
- ✓ **Complexity:** Robustly handles non-linear feature interactions.

RQ2: REGRESSION

Best: XGBoost Regressor

Modeling Total Expenditure Drivers

Models evaluated: Regularized Regressions, Tree-based Ensembles

- ✓ **Max Variance:** Achieved R^2 of 0.968, explaining nearly all variance.
- ✓ **Accuracy:** Recorded lowest MAE (≈ 480) and RMSE (≈ 1300).
- ✓ **Behavioral Fit:** Captures diminishing marginal effects of income.

RQ3: CLUSTERING

Best: KMeans (k=7)

Consumer Segment Identification

Models evaluated: KMeans (various k), Hierarchical Clustering

- ✓ **Cluster Quality:** Highest Silhouette Score (0.388) at k = 7.
- ✓ **Stability:** Optimal balance of intra-cluster similarity and separation.
- ✓ **Interpretability:** Provides a robust framework for profiling behavior.

ENSEMBLE PERFORMANCE

Best: Soft Voting

Maximizing Predictive Robustness

Models evaluated: Hard Voting, Soft Voting, Stacking

- ✓ **Error Trade-off:** Highest F1-score (0.68) vs. Stacking/Hard Voting.
- ✓ **Reliability:** Lowest false-positive rate among ensemble methods.
- ✓ **Discrimination:** Strong ROC-AUC (0.983) for effective ranking.

THANK YOU